

Chapter 1 Data exploration

Students should be able to summarise datasets and construct charts.

Learning objective: Describe measures of central tendency.

Learning objective: Construct histograms and boxplots.

Chapter 2 Probability

Students should apply probability rules to actuarial problems.

Learning objective: Apply Bayes theorem to update beliefs.

Learning objective: Compute conditional probabilities.

Chapter 3 Distributions

Students should identify and use common probability distributions.

Learning objective: Select appropriate discrete models.

Learning objective: Select appropriate continuous models.

Chapter 4 Inference

Students should estimate parameters and test hypotheses.

Learning objective: Construct confidence intervals.

Learning objective: Interpret p-values carefully.

Chapter 5 Regression

Students should fit and critique linear models.

Learning objective: Interpret slope and intercept.

Learning objective: Diagnose residual patterns.

Chapter 6 Credibility

Students should apply credibility methods in pricing contexts.

Learning objective: Compute classical credibility premiums.

Learning objective: Compare classical and Bayesian approaches.